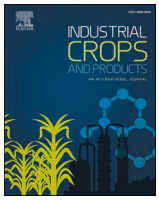
Industrial Crops & Products 187 (2022) 115157



Contents lists available at ScienceDirect

Industrial Crops & Products

journal homepage: www.elsevier.com/locate/indcrop





An integrated olive stone biorefinery based on a two-step fractionation strategy

Carmen Padilla-Rascón [a,b], Florbela Carvalheiro [c], Luís C. Duarte [c], Luisa B. Roseiro [c], Encarnación Ruiz [a,b,*], Eulogio Castro [a,b]

[a] Department of Chemical, Environmental and Materials Engineering, Universidad de Jaén, Campus Las Lagunillas, Jaén 23071, Spain
[b] Center for Advanced Studies in Earth Sciences, Energy and Environment (CEACTEMA), Universidad de Jaén, Campus Las Lagunillas, Jaén 23071, Spain
[c] National Laboratory of Energy and Geology, Bioenergy and Biorefineries Unit, Estrada do Paço do Lumiar 22, Lisboa 1649-038, Portugal

ARTICLE INFO

Keywords:
Bioethanol
Biomass pretreatment
Ethanol-water organosolv
Furfural
Lignin-derived products
Olive oil industrial waste

ABSTRACT

Olive stones (OS) constitute a waste lignocellulosic material produced by the olive oil industry in great amounts, that currently is only used as a low-value energy source for industrial or domestic boilers. Having in view its full valorization, this work proposes and validates an integrated strategy aiming to obtain three different streams of sugars / lignin-derived compounds. Dilute acid hydrolysis was used to obtain a xylose-rich hydroysate that was chemically converted into furfural with a 48.7 % yield. The resulting acid-pretreated solid biomass that consisted mainly of lignin and cellulose, was subjected to a catalyzed ethanol-based organosolv delignification. Temperature, time, and sulphuric acid concentration were optimized in order to recover added-value lignin products and digestible cellulose.

At the optimal conditions (190 °C and 30 min), a 50 % delignification was reached, together with the highest enzymatic hydrolysis yields (190 g glucose/kg of OS). Phenolic compounds content in organosolv liquors reached 41.6 mg GAE/g OS. This extract presented an antioxidant capacity up to 10.9 mg TE/g OS. The pretreated solid fraction was used as a substrate for ethanol production by a pre-saccharification and simultaneous saccharification and fermentation process, enabling to obtain an ethanol concentration of 47 g/L, with a fermentation yield of 61.4 % of the theoretical maximum. Globally, from 100 kg of OS processed according to this experimental scheme, 6.9 kg of furfural, 6.2 kg of ethanol, 7.4 kg of lignin, and 4.2 kg of phenolics compounds can be obtained as main products, thus constituting a way of valorization of renewable material in a multiproduct biorefinery strategy.

1. Introduction

Lignocellulosic biomass (LCB) has a high potential for the production of fuels and value-added chemicals, in a biorefinery concept (Scopel and Rezende, 2021). The production of bioethanol and petroleum-derived products from residual LCB is a sustainable way to reduce dependence on fossil fuels, due to their abundance, renewability, and non-competition with food (Ferreira and Taherzadeh, 2020). One example of such lignocellulosic biomass are the olive stones.

Olive stones (OS) are a LCB generated in the olive oil production process, where other olive-derived residual biomass is produced, making this agroindustrial sector an interesting source of raw materials in a biorefinery context (Ruiz et al., 2017). To obtain olive oil, the whole olive is subjected to milling as a first step, then a malaxing process, and finally the separation of the oil from the rest of the mixture (olive pomace). The olive pomace is further conducted to an extraction facility to obtain olive pomace oil by solvent extraction and a final solid residue called exhausted olive pomace (EOP).

In recent years, the separation of the OS from the milled olives or from the final olive pomace has become frequent practice, due to its use as fuel in small industrial and domestic boilers (Manzanares et al., 2017), but still, this is a niche market. In order to illustrate the volume of available olive stones, and considering only Spain, as the main olive-producing country in the world, an estimation of 750,000 tonnes of OS is considered, based on the fact that it produces about 7.5 million tonnes of olive fruit in 2021 (https://www.fao.org/faostat/en/), and assuming that OS accounts for approximately 10 % by weight of the olive fruit (Romero-García et al., 2014). Moreover, OS has the advantage

* Corresponding author at: Department of Chemical, Environmental and Materials Engineering, Universidad de Jaén, Campus Las Lagunillas, Jaén 23071, Spain.
E-mail addresses: eruiz@ujaen.es, eruiz@ujaen.es (E. Ruiz).

https://doi.org/10.1016/j.indcrop.2022.115157
Received 3 February 2022; Received in revised form 15 May 2022; Accepted 27 May 2022
Available online 20 July 2022
0926-6690/© 2022 Elsevier B.V. All rights reserved.

of not needing to be milled for its use and of being located in the olive mills and olive pomace extracting industry, reducing costs and facilitating operational logistics.

Chemically, OS mainly contains lignin (35.6 %), followed by hemicellulose (26 %), and cellulose (20.9 %) (Padilla-Rascón et al., 2020b), which makes it suitable for valorization in the context of a LCB biorefinery. However, its complex structure and recalcitrance makes it difficult to take full advantage of the different fractions contained in OS.

Multiple pre-treatments have been reported to overcome the recalcitrant structure of LCB by breaking the polymeric bonds (Bhatia et al., 2020). In general, more severe conditions are required for the recovery of cellulose and lignin than those used for the recovery of the hemicellulosic fraction, since it is a more labile fraction and degrades more easily (Huijgen et al., 2012; Romero et al., 2015; Zhao et al., 2009). Therefore, a two-stage pre-treatment is preferred to recover both hemicellulose and cellulose sugars with adequate yields (Padilla-Rascón et al., 2020b).

Dilute acid and liquid hot water pre-treatments are two widely used pre-treatment methods for the hydrolysis of the hemicellulosic fraction (Yang et al., 2020). In previous work on OS, the conditions to maximise the recovery of hemicellulosic sugars in the liquid fraction were optimized (128 °C, 10.5 g acid/100 g OS and 33 % solids) leaving a pre-treated solid that consists mainly of cellulose and lignin (Padilla-Rascón et al., 2020b). Lignin is often considered a physical barrier blocking the access of enzymes to cellulose hydrolysis which binds to cellulase complex in a non-productive way and deactivates it (Miliotti et al., 2019). The use of organosolv pre-treatments aims at delignification together with partial hydrolysis of hemicellulose and a reduction in the degree of polymerisation and crystallinity of the cellulose fraction (Mosier, 2005; Taherzadeh and Karimi, 2008). This makes the cellulose fibrils more accessible to enzymes and increases the yields of enzymatic saccharification (Huijgen et al., 2012; Pan et al., 2005). In addition for increasing enzymatic saccharification, organosolv pre-treatment has the advantage of producing high purity lignin streams (Huijgen et al., 2012), which is of relevance, as lignin is a potential source for obtaining aromatic chemicals, such as phenolic compounds with antioxidant properties (Zakzeski et al., 2010). The most promising organosolv solvents are ethanol, formic acid, and acetic acid. In particular, the most studied and used is ethanol, as it can solubilise lignin, has low toxicity, high volatility, is easily recyclable, cheap and green (Chen et al., 2015).

However, due to the high recalcitrance of OS, the delignification of this biomass has been scarcely studied as the data obtained were not very promising. Thus, to avoid biomass recalcitrance and to take full advantage of biomass composition, process configurations involving a two-stage pre-treatment, in which a second organosolv stage is employed have been studied for different LCB, such as rice straw (Moniz et al., 2015), wheat straw (Huijgen et al., 2012) and grape stalks (Amendola et al., 2012).

The aim of this work was to study the effectiveness of organosolv pre-treatment in an integrated OS fractionation strategy, in order to optimise the recovery conditions of sugars (mainly glucose and xylose) and lignin. The proposed scheme involved a first dilute acid pre-treatment step for the recovery of xylose, which was used for the production of furfural. The second organosolv pre-treatment step aimed to improve the cellulose saccharification potential by lignin solubilization. The recovery of lignin and bioactive phenolic compounds in the liquid fraction has also been addressed. Ethanol production from the cellulose enriched solid by a pre-saccharification and simultaneous saccharification and fermentation process (PSSF) has been evaluated.

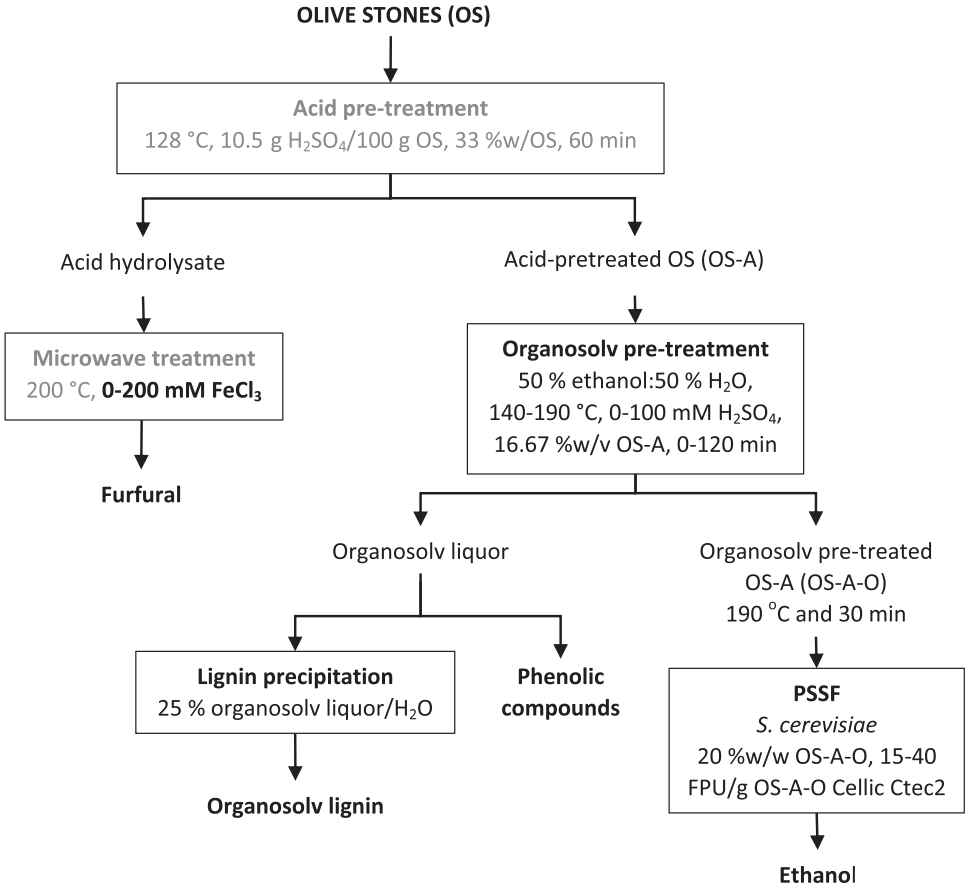


Fig. 1. Scheme summarizing the strategy used for the valorisation of the main OS fractions. In grey, is the experimental procedures optimised in previous works (Padilla-Rascón et al., 2020b, 2021).

2. Material and methods

The strategy to upgrade OS-A here developed is detailed in Fig. 1. It comprises a two-step fractionation approach to enable the full upgrade of both the polysaccharides and the lignin fractions.

2.1. Raw material

Shattered olive stones (OS) were kindly provided by a local company in Jaén (Andalucía, Spain). They had a moisture content of 8 % and a particle size between 0.1 and 0.32 cm. OS were subjected to acid hydrolysis pre-treatment in an autoclave, under previously optimized conditions for efficient solubilisation of the hemicellulosic fraction (128 °C, solid/liquid ratio of 33 % and 10.5 g H_2SO_4/100 g OS, for 60 min) (Padilla-Rascón et al., 2020b). The process was performed in 1 L-ISO bottles filled with 500 mL solution and 165 g dry OS, and multiple repetitions were performed to obtain enough amount of pre-treated material. The resulting slurry was vacuum filtered to separate the liquid from solid fractions. The xylose enriched liquid fraction was chemically characterised as described in Section 2.7.2 and stored at 4 °C, then subsequently used to obtain furfural. The solid fraction was washed with distilled water, dried at 40 °C, mixed to homogenise, chemically characterised (according to Section 2.7.1) and stored at room temperature for further use. The solid fraction obtained in this acid pre-treatment (OS-A) formed mainly by cellulose and lignin, was the biomass used as raw material to be fractionated by organosolv pre-treatments in the present work.

2.2. Organosolv pre-treatment of OS-A

The organosolv treatment was carried out in a 600 mL Parr reactor (482 model), using OS-A as feedstock, and an ethanol/water mixture (50:50, w/w) as usually described before, e.g. (Alves-Ferreira et al., 2021; Wildschut et al., 2013). The solid/liquid ratio used was 16.7 %. The previous conditions were fixed for all experiments, but the effects of temperature (140, 165, and 190 °C), sulphuric acid concentration (0, 50 or 100 mM), and reaction time (0–120 min) were studied.

Upon the reaction completion, the reactor was rapidly cooled down to 50 °C and the solid and liquid fractions were separated by vacuum filtration. The solid fraction (OS-A-O) was washed with the same volume of ethanol/water solution used in the organosolv pre-treatment and subsequently, with twice the volume of distilled water. When required, the solid pretreated biomass was dried at 40 °C and stored at room temperature. Pretreated biomass was analysed according to Section 2.7.1. The liquid fraction was analysed according to Section 2.7.2 and stored at 4 °C.

2.3. Lignin recovery

The delignification yield in the solid fraction after organosolv pre-treatment was calculated from the acid-insoluble lignin (AIL) content in the pretreated OS before (OS-A) and after organosolv pre-treatment (OS-A-O). AIL is considered to be the acid-insoluble fraction obtained after the solid characterization (Section 2.7.1) (Eq. (1)). Calculations were performed according to (Moniz et al., 2015).

$$\text{Delignification yield}\ (\%) = 100 - \text{organosolv solid yield}\ (\%) \cdot \frac{\text{AIL OS} - \text{A} - \text{O (g)}}{\text{AIL OS} - \text{A (g)}} \quad (1)$$

Solubilised lignin in the liquor obtained from the organosolv treatment was precipitated by diluting the hydrolysates with water (4:1, water:liquor ratio w/w), incubated in an orbital shaker (Comecta, Spain) for 2 h at 30 °C, and then the mixture was separated by centrifugation (6000 rpm) for 20 min at room temperature (centrifuge Ortoalresa, Digicen 21 R, Spain) and the supernatant was separated from the precipitated material by decanting. The recovered lignin precipitate was dried at 40 °C for 48 h and quantified by weighing (Eq. (2)). This procedure was carried out in triplicate. An overall recovered lignin yield after organosolv pre-treatment was calculated (Eq. (3)) taking into account the lignin recovered in the solid fraction, the precipitated lignin, and the total phenolic compounds determined as described in Section 2.7.3:

$$\text{Precipitated lignin yield}\ (\%) = \frac{\text{Precipited AIL OS} - \text{A} - \text{O (g)}}{\text{AIL OS} - \text{A (g)}} \times 100 \quad (2)$$

$$\text{Overall recovered lignin yield}\ (\%) = \frac{\text{AIL OS} - \text{A} - \text{O (g)} + \text{Precipited AIL OS} - \text{A} - \text{O (g)} + \text{Total phenols (g)}}{\text{AIL OS} - \text{A (g)}} \times 100 \quad (3)$$

2.4. Enzymatic hydrolysis

The solids obtained in each pre-treatment (OS-A and OS-A-O) were used as a substrate for enzymatic hydrolysis (EH) to establish the effect of pre-treatment conditions on enzymatic saccharification, using a method described before (Padilla-Rascón et al., 2020b). Briefly, in a 100 mL Erlenmeyer flask the substrate (5%w/v), was mixed with sodium citrate buffer 0.05 M pH = 4.8 and Cellic® CTec2 (Novozymes A/S, Bagsværd, Denmark), with a concentration of 15 FPU/g of dry substrate supplemented with β-glucosidase (Novozymes 50010), 10 % of Cellic® CTec2. The mixture was stirred at 150 rpm in an orbital shaker at 50 °C for 72 h. Each assay was performed in triplicate. Samples of EH liquid were taken every 24 h and analysed by HPLC as detailed in Section 2.7.2. The saccharification yield, and the yield of the EH are calculated according to Eq. (4) and Eq. (5), respectively:

$$\text{Saccharification yield}\ (\%) = \frac{\text{g glucose by EH}}{\text{g glucose in OS} - \text{A} - \text{O}} \times 100 \quad (4)$$

$$\text{EH yield}\ (\%) = \frac{\text{g glucose by EH}}{\text{g glucose in OS} - \text{A}} \times 100 \quad (5)$$

2.5. Pre-sacchararification and simultaneous saccharification and fermentation (PSSF) of OS-A-O

After selecting the organosolv condition with the highest EH yield (190 °C and 30 min), the OS-A-O obtained was subjected to a pre-saccharification followed by simultaneous saccharification and fermentation (PSSF) process. The fermenting microorganism was *Saccharomyces cerevisiae* (Fermentis ethanol red, France). The inoculum was grown in an orbital shaker at 30 °C and 150 rpm for 24 h in a culture medium composed of: 2 g/L NH_4Cl; 1 g/L KH_2PO_4; 0.3 g/L $MgSO_4.7\ H_2O$ and 30 g/L glucose. PSSF experiments were performed in 100 mL Erlenmeyer flasks, with 20 mL of 0.05 M sodium citrate buffer and 20 %w/w OS-A-O. The enzyme Cellic® CTec2 loadings were 15, 30 and 40 FPU/g OS-A-O (Novozymes A/S, Bagsværd, Denmark) and

supplemented with β-glucosidase (Novozymes 50010) at a 10 % of Cellic® CTec2 loading.

The pre-saccharification step lasted 4 h at 50 °C and 150 rpm. After that, the inoculum was added at a cell concentration of 0.25 g/L, the temperature was adjusted to 35 °C and shaken at 150 rpm for 72 h. Samples were taken for analysis in 24 h periods to determine glucose and ethanol concentrations, as described in Section 2.7.2. All assays were performed, at least, in triplicate, and average results are shown.

2.6. Production of furfural from acid hydrolysate

The xylose-rich hydrolysate obtained in the first acid pre-treatment step under the optimal conditions previously described (128 °C, 10.5 g H_2SO_4/100 g OS, and solid/liquid ratio of 33 %, according to Padilla-Rascón et al., 2020b), was used for the production of furfural.

The experimental procedure was carried out according to the conditions optimised before (Padilla-Rascón et al., 2021). The experiments were carried out in a microwave reactor (Anton Paar Monowave 400, Graz, Austria), using 10 mL capped glass vessels loaded with 4 mL of hydrolysate, and the temperature was set at 200 °C. The catalysts used for the dehydration of xylose into furfural were H_2SO_4 and $FeCl_3$. Sulphuric acid is already present in the hydrolysate due to the acid pre-treatment, hence, no further addition is required. Iron chloride was added in different concentrations (0–200 mM) to determine the amount required to maximize furfural yields. The liquors obtained were analysed according to Section 2.7.2. The experiments were performed in triplicate and average results are shown.

2.7. Analytical methods

2.7.1. Chemical characterisation of solids

Raw OS and the solids obtained in the pre-treatments (OS-A and OS-A-O) were characterised according to the National Renewable Energy Laboratory (NREL) methodology (Sluiter et al., 2012). Briefly, for the characterization of the solids, two-stage acid hydrolysis was performed, the first stage at low temperature (30 °C) and high concentration of H_2SO_4 (72 %), followed by another acid stage at high temperature (120 °C) and low concentration of H_2SO_4 (4 %). Solid and liquid fractions were separated by vacuum filtration, and the liquid fraction was analysed by HPLC (Section 2.7.2) and measured spectrophotometrically to determine its acid soluble lignin concentration (ASL). The resulting solid fraction was washed, dried and weighed, and muffled (550 °C, for at least 5 h) to evaluate acid insoluble lignin (AIL). Ash content of OS, OS-A and OS-A-O was also determined using the same procedure.

The raw OS was subjected to two previous extractions, first with water (24 h) and then with ethanol (24 h) for extractives. Replicates of all determinations were carried out in triplicate, at least.

2.7.2. Chemical characterisation of liquors

All the liquid fractions obtained from pre-treatments, enzymatic hydrolysis, furfural and ethanol production processes, as well as the hydrolysates generated during OS-A and OS-A-O characterization, were analysed by high performance liquid chromatography (HPLC). For the determination of the monomeric sugars (glucose, xylose, galactose, and arabinose) an HPLC in a Waters Prostar liquid chromatograph with a refractive index detector (Waters 2414), equipped with a CARBOSep CHO-782 Pb column was used, operating at 70 °C with ultrapure water as mobile phase (0.6 mL/min), being the samples previously neutralized with $CaCO_3$. Ethanol, furfural, 5-hydroxymethylfurfural (HMF), acetic, levulinic and formic acid were determined by HPLC using an Agilent Technologies liquid chromatograph (1260 model) with the ICSep ICE-COREGEL 87H3 column operating at 65 °C with 5 mM sulphuric acid as mobile phase (0.6 mL/min). Liquors obtained in organosolv pre-treatment and ethanol and furfural production processes were analysed completely with the ICSep ICE-COREGEL 87H3 column. All samples were filtered before analysis through 0.22 μm nylon filters.

2.7.3. Quantification and identification of total phenolic compounds

Total phenolic compounds (TPC) were measured spectrophotometrically, following the Folin-Ciocalteu method (Singleton and Rossi, 1965). Some modifications were made, namely, 0.3 mL of diluted extract was added to 2.5 mL of Folin-Ciocalteu solution (10%). Subsequently, 2 mL of Na_2CO_3 solution (7.5%w/v) was added and the mixture was homogenised, and stored in the dark for 1 h after which the absorbance was measured at 750 nm in a Bio-Rad iMarkTM microplate absorbance reader (Hercules, CA, USA). Gallic acid was used as a reference standard compound. Results are expressed as Gallic Acid Equivalents (GAE). Analyses were performed in triplicate and average results are shown.

The determination of PVPP (polyvinylpolypyrrolidone) precipitable phenolic compounds was performed by their adsorption on the insoluble PVPP matrix. The experimental procedure was carried out according to Roseiro et al. (2013). Briefly, 100 mg of PVPP, 1 mL of acidified distilled water (pH 3) and 1 mL of organosolv liquor were added to Eppendorf tubes. They were allowed to stand for 15 min at 4 °C and then centrifuged for 10 min. The supernatant was used to determine the non-adsorbed compounds following the method described above. The values obtained were subtracted from the total phenol content to obtain the total value of phenolics precipitated by PVPP. Experiments were carried out in triplicate.

The liquors obtained from the organosolv treatment were analysed for their phenolic profile using a capillary electrophoresis system (Agilent Technologies, Waldbronn, Germany) equipped with a diode array detector, according to the methodology described in Roseiro et al. (2013). Detection was performed at 320 nm and compounds were identified by comparison with UV spectra and migration times of phenolic standards analysed under the same conditions as liquors.

2.7.4. Determination of antioxidants activity

The antioxidant activity of the liquors was quantified by a spectrophotometric method using DPPH (2,2-diphenyl-1-picrylhydrazyl) as described in (Gómez-Cruz et al., 2021). Briefly, 2 mL of a solution containing methanol and 6×10^{-5} M DPPH (2,2-diphenyl-1-picrylhydrazyl) was mixed to 200 μL of the diluted sample, shaken, and left to stand for 15 min in the dark. Absorbance was then measured at 490 nm in a Bio-Rad iMarkTM microplate absorbance reader (Hercules, CA, USA). Trolox was used as a reference standard compound. Analyses were performed in triplicate and average results are shown.

The antioxidant activity of the liquors was expressed as Trolox equivalents (TE) and described by the DPPH percentage inhibition, determined as reported before (Roseiro et al., 2013).

3. Results and discussion

3.1. Feedstock composition

3.1.1. Raw olive stones (OS)

Original OS have a significant content of polysaccharides, 21.0 % cellulose and 24.8 % hemicellulose, with xylose being 86 % of the total hemicellulose. Compared with other olive-derived biomass (Contreras et al., 2020), OS are the olive by-product with the highest polysaccharide content, i.e., 46 % vs 36 % in olive tree pruning (OTP) or 21 % and 16 % extracted olive pomace (EOP) and olive leaves (OL), respectively. Most noteworthy is also the OS lignin content, which reaches 32.9 %, considerably higher than the reported for OTP (18 %), OL (19 %), and EOP (22 %). This high lignin content may explain the high recalcitrancy of OS that makes it one of the most difficult industrial olive derived wastes to hydrolyse (Doménech et al., 2020) In contrast to other materials, OS extracts content (4.1 %) is much lower than the values found in the rest of olive-derived biomasses (29 %, 39 % and 49 % for OTP, OL, and EOP, respectively). In fact, previous works on valorization processes of these olive-derived biomasses usually involved a first extraction step to remove most of the extractive fraction before applying

different pre-treatment strategies to release sugars from the hemicellulose and cellulose fractions (López-Linares et al., 2020, 2018; Martínez-Patiño et al., 2017). Therefore, in the case of OS, no extraction is needed before pre-treatment.

3.1.2. Acid pre-treated olive stones (OS-A)

In this work, a fractionation strategy was applied involving a first acid pre-treatment step previously optimized to solubilize hemicellusoses followed by an organosolv delignification process, as described in Section 2.1. The first acid stage allows the hydrolysis of the hemicellulosic fraction and the recovery of xylose-rich hydrolysate. If the organosolv pre-treatment were carried out directly on biomass, this fraction would be lost by degradation as it would undergo the more severe conditions necessary to solubilise lignin and obtain a solid susceptible to saccharification (Huijgen et al., 2012).

Table 1 also shows the composition of OS-A. The acid pre-treatment solid yield was 63.1 %. This biomass is mainly composed of cellulose and lignin, as both fractions were retained in OS-A in a percentage of 95 %, while more than 80 % of the hemicellulosic fraction was solubilized. This solid fraction was the biomass used in this work as raw material for all organosolv pre-treatment experiments described below. The composition of the liquid fraction rich in xylose and its transformation into furfural are described in Section 3.5.

3.2. Organosolv fractionation of OS-A biomass

3.2.1. Effect on biomass composition

The main goal for the utilization of the organosolv pre-treatment is biomass delignification. Typically, the main used solvent is ethanol, and although ethanol-organosolv delignification can be carried out using ethanol concentrations ranging from 40 % to 75 % (Zhang et al., 2016). Typical values are usually in the range 50–60 % (Huijgen et al., 2011, 2010), with 50 % being particularly reported.

Delignification enables also to increase the cellulose content of the pre-treated solids and the enzymatic digestibility of the cellulose fraction, together with the possibility to recover added-value lignin-derived products, as this pre-treatment allows to obtain these in a liquid stream. The organosolv pre-treatment conditions and the composition of the solids obtained (OS-A-O) are shown in Table 2, revealing the composition of OS-A-O solids obtained under the diverse conditions tested. A significant increase in cellulose content was observed as compared to the OS-A composition. Hemicellulose was completely solubilised during organosolv pre-treatment, except for a single condition (190 °C and 0 min), where a residual xylose content remained. Lignin content was also decreased significantly.

Focusing on the influence of the different experimental conditions tested on biomass fractionation, a significant effect of temperature, acid concentration, and time was observed. In the three first experiments performed at a lower temperature (140 °C), OS-A-O cellulose content was found to increased at a statistical significant confidence level of 95 % with the increase in sulphuric acid concentration, while the lignin content decreased (Table 2). Under these conditions, there was apparently no loss of glucose in the solid and the degree of delignification increased with the addition of acid up to 37 % (Table 3). A notable effect of temperature was observed when it was increased from 140 to 165 °C, both in the experiments carried out with and without acid. At 165 °C part of the cellulose is solubilised (up to 20 %), but an increased delignification is achieved, surpassing 40 %.

In the experiments conducted at the higher temperature (190 °C) no acid was added, as at lower temperatures cellulose degradation was already observed with the addition of acid. The effect of different pre-treatment times was studied at 190 °C. The cellulose content in OS-A-O rose with increasing time up to 30 min (maximum of 54.5 % cellulose). At longer times (60, 90, and 105 min), the cellulose content decreased with time while the lignin content increased. Glucose recovery in the solid decreased with increasing time, with losses of more than one-third of the cellulose after 60 min. The highest levels of delignification were obtained at 30 and 60 min, reaching delignification percentages of around 50 %. This value is comparable to the same obtained by (Gómez-Cruz et al., 2022) with another olive by-product, EOP, reaching a delignification of 53 % with an organosolv pre-treatment at 140 °C, 60 min and 1 % H_2SO_4 after a two-step aqueous extraction. Slightly higher delignification (66 %) was achieved with olive tree pruning after organosolv pre-treatment at 66 % ethanol, 210 °C, 60 min (Díaz et al., 2011). Other biomasses, such as wheat straw, have been pre-treated with the same strategy as in this work, (an acid pre-treatment followed by an organosolv pre-treatment), obtaining a maximum delignification of 69 % (Huijgen et al., 2012).

To the best of our knowledge, this is the first work describing ethanol organosolv delignification of OS, with a relevant delignification yield (up to 52.5 %) and no catalyst added.

3.2.2. Effect on hydrolysate composition

The composition of the organosolv liquors obtained is shown in Table 4. Sugar concentration detected, ranged from low values (3.1 g/L) to moderate values (13.3 g/L). Xylose and arabinose come from the solubilization of the remaining hemicellulosic fraction still present in OS-A and are the most relevant sugars. Glucose come from cellulose losses during pre-treatment (as can be seen in Table 3), especially in experiments performed at 190 °C for longer reaction times. Besides sugars, other compounds such as acetic, formic and levulinic acids, or furanic compounds, were also detected (Table 4). Apart from acetic acid, this group of compounds comes from sugar degradation (Jönsson et al., 2013), mainly under more severe conditions. For example, the addition of 50 mM of sulphuric acid at 165 °C generated higher furan concentrations (2.1 g/L of HMF and 4.1 g/L of furfural) than in the experiment carried out at the same temperature without acid and then in the experiment performed at 140 °C and 50 mM acid. On the other hand, longer pre-treatment times at 190 °C generated higher concentrations of degradation products, reaching 10.4 g/L of furanic compounds and 5.9 g/L of acids at 105 min.

Due to the low sugars concentrations and high concentrations of degradation products that can act as fermentation inhibitors, these liquors are not suitable for ethanol production (Kim, 2018). As such, the valorisation of these liquid streams was preferably considered for the recovery of lignin-derived compounds, as described in the following subsections.

3.2.2.1. Phenolic compounds and antioxidant activity. Liquid fractions obtained from organosolv pre-treatment have also been analysed to determine the concentration of total phenolic compounds (TPC), phenolic compounds precipitable by PVPP (PP PVPP), and antioxidant capacity (Table 5).

As can be seen in Table 5, TPC and PP PVPP showed the same trend at 140 °C, both increasing with the addition of acid. The highest values at

Table 1
Composition of raw olive stones (OS) and acid pre-treated olive stones (OS-A). Values are expressed as a percentage of dry weight.

Composition	Raw OS	OS-A
Cellulose	21.0	31.7
Hemicellulose	24.8	7.3
Xylose	24.0	5.2
Galactose	2.0	1.8
Arabinose	2.0	1.3
Acetyl groups	3.3	0.7
Lignin	32.9	49.5
AIL	31.7	48.3
ASL	1.2	0.8
Ash	0.29	0.7
Extracts	4.1	–

AIL: Acid insoluble lignin; ASL: Acid soluble lignin. Deviations < 5 %.

Table 2
Composition of the solids obtained after organosolv pre-treatment (OS-A-OL) in the different experimental conditions assayed. Values are expressed as a percentage of dry weight.

Sample	Temperature (°C)	H_2SO_4 (mM)	Time (min)	Cellulose	Hemicellulose	Total lignin	AIL	ASL	Ash
OS-A-O_140	140	–	120	46.6	0	47.1	46.7	0.35	0.26
OS-A-O_140-A50	140	50	120	49.6	0	46.1	45.7	0.40	0.15
OS-A-O_140-A100	140	100	120	53.8	0	44.2	43.9	0.26	0.42
OS-A-O_165	165	–	120	48.7	0	42.8	42.6	0.27	0.07
OS-A-O_165-A50	165	50	120	46.8	0	46.3	46.0	0.38	0.26
OS-A-O_190–0	190	–	0	50.7	1.49	44.5	44.1	0.46	0.09
OS-A-O_190–30	190	–	30	54.5	0	44.5	44.0	0.49	0.19
OS-A-O_190–60	190	–	60	46.4	0	44.8	44.5	0.27	0.05
OS-A-O_190–90	190	–	90	45.9	0	57.4	57.0	0.37	0.09
OS-A-O_190–105	190	–	105	31.9	0	64.5	64.3	0.18	0.02

AIL: Acid insoluble lignin; ASL: Acid soluble lignin. Deviations < 5 %.

Table 3
Cellulose and lignin yields obtained in the solids after organosolv pre-treatment (OS-A-O).

Sample	Solid yield[a] (%)	Cellulose yield[b] (%)	Delignification yield[c] (%)
OS-A-O_140	79.2	100	26.3
OS-A-O_140-A50	72.3	99.1	26.7
OS-A-O_140-A100	69.6	98.9	37.0
OS-A-O_165	64.4	86.7	43.6
OS-A-O_165-A50	61.4	79.4	41.9
OS-A-O_190–0	74.2	100	32.6
OS-A-O_190–30	52.4	78.9	49.8
OS-A-O_190–60	51.8	66.4	52.5
OS-A-O_190–90	48.3	61.2	43.4
OS-A-O_190–105	45.7	40.3	39.5

Deviations < 5 %.
[a] Solid yield: expressed as g OS-A-O /100 g OS-A.
[b] Cellulose yield: cellulose contained in OS-A-O, as a percentage of cellulose contained in OS-A.
[c] Delignification yield: AIL removal in OS-A-O, as a percentage of AIL contained in OS-A.

this temperature were obtained with the addition of 100 mM H_2SO_4 (11.3 g GAE / 100 g OS-A lignin of TPC and 6.0 g GAE / 100 g OS-A lignin of PP PVPP). This trend was also observed in the concentration of phenols obtained in the tests carried out at 165 °C, reaching the highest concentrations with the addition of 50 mM H_2SO_4 (15.6 g GAE / 100 g OS-A lignin of TPC and 7.3 g GAE / 100 g OS-A lignin of PP PVPP). At 190 °C, no notable differences were detected at the different pre-treatment times, with TPC ranging from 9.4 to 14.3 g GAE / 100 g OS-A lignin and PP PVPP ranging from 3.5 to 6.8 g GAE / 100 g OS-A lignin. The maximum concentration of TPC obtained was 4.7 kg GAE/ 100 kg OS, exceeding the value was obtained from organosolv delignification of EOP, 2.4 kg GAE/ 100 kg OS (Gómez-Cruz et al., 2022).

Table 5 also shows the antioxidant capacity of the liquors. The results are in the range 2.7–3.6 g TE / 100 g OS-A lignin, without notable variability with the experimental conditions used during organosolv pre-treatment. DPPH inhibition was below 75 %.

The liquors generated in the organosolv pre-treatment were also analysed by capillary electrophoresis to determine their phenolic profile and identify the main phenolic compounds present. Complex matrices were obtained, as can be seen in Fig. 2 where the electropherogram of the organosolv pre-treatment performed at 190 °C and 30 min is shown. As can be seen, the identified compounds are putatively derived from the lignin fractionation, except for furfural, also identified by HPLC in the organosolv liquors (Table 4), that is derived from sugars (pentoses). Besides, the identified compounds there are several other compounds that were not yet identified. These are probably also lignin-derived compounds, for instances, catechol or its derivatives, as indicated by

Table 5
Total concentrations of phenolic compounds (TPC) (g GAE/100 g OS-A lignin), phenols precipitated by PPVP (PP PVPP) (g GAE/100 g OS-A lignin), antioxidants capacity (g TE/100 g OS-A lignin) and DPPH inhibition (%) of the liquid streams obtained in the first acid pre-treatment and the second organosolv pre-treatment at the different experimental conditions assayed.

Sample	TPC	PP PVPP	g TE/100 g	% DDPH inhibition
Acid-pretreated OS	2.8	1.4	2.1	50.9
OS-A-O_140	6.1	3.0	2.7	51.3
OS-A-O_140-A50	9.1	4.3	3.4	63.3
OS-A-O_140-A100	11.3	6.0	3.5	64.3
OS-A-O_165	12.5	5.4	3.4	63.5
OS-A-O_165-A50	15.6	7.3	3.5	64.1
OS-A-O_190–0	9.4	3.7	2.8	53.2
OS-A-O_190–30	13.6	6.7	3.6	73.5
OS-A-O_190–60	14.1	4.7	3.6	68.2
OS-A-O_190–90	14.3	6.8	3.6	65.7
OS-A-O_190–105	13.6	3.5	3.5	64.7

Deviations < 5 %.

Table 4
Composition of the liquors obtained in the organosolv pre-treatment.

Sample	Glucose (g/L)	Xylose (g/L)	Arabinose (g/L)	Formic acid (g/L)	Acetic acid (g/L)	Levulinic acid (g/L)	HMF (g/L)	Furfural (g/L)
OS-A-O_140	0.2	4.8	0.37	2.9	0.01	0.00	0.01	0.68
OS-A-O_140-A50	1.1	4.7	1.2	3.3	0.56	0.02	0.11	1.6
OS-A-O_140-A100	1.5	4.3	1.6	5.1	0.48	0.01	0.21	2.4
OS-A-O_165	1.8	3.7	1.9	1.1	0.52	0.05	0.57	3.2
OS-A-O_165-A50	0.58	1.2	1.3	1.7	1.2	0.48	2.1	4.1
OS-A-O_190–0	1.1	5.5	1.0	1.7	0.42	0.00	0.08	0.94
OS-A-O_190–30	3.7	3.3	1.8	1.8	0.93	0.13	1.8	4.1
OS-A-O_190–60	2.7	2.2	2.2	1.8	1.3	0.28	2.8	4.4
OS-A-O_190–90	4.8	2.8	2.6	2.3	1.1	0.83	4.7	5.1
OS-A-O_190–105	6.4	3.2	3.8	2.4	1.2	2.3	5.1	5.3

Deviations < 5 %.

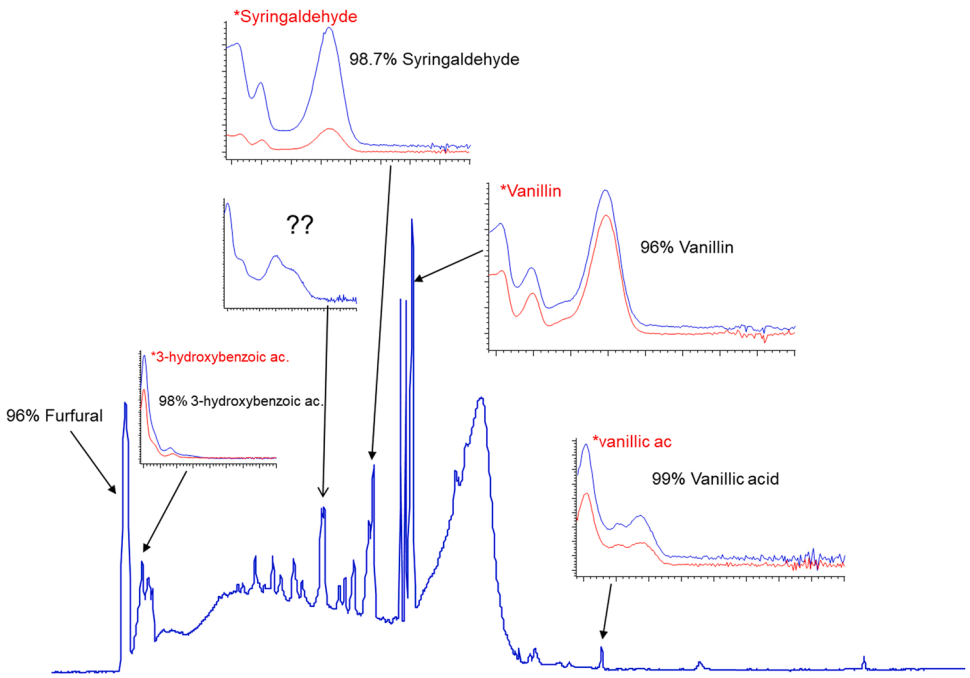


Fig. 2. Electropherogram (320 nm) showing the phenolic profile of the soluble lignin products obtained after organosolv pre-treatment. Match percentage was obtained by comparison with the authentic standards analysed under the same conditions.

Table 6
Lignin recovered and lignin precipitated after organosolv delignification.

Sample	Precipitated lignin yield[a]	Overall recovered lignin[b]
OS-A-O_140	7.8	90.5
OS-A-O_140-A50	19.5	97.0
OS-A-O_140-A100	21.7	96.4
OS-A-O_165	20.5	89.8
OS-A-O_165-A50	18.0	92.1
OS-A-O_190–0	14.4	91.6
OS-A-O_190–30	19.5	80.9
OS-A-O_190–60	19.7	81.5
OS-A-O_190–90	20.0	87.6
OS-A-O_190–105	20.0	93.3

Deviations < 5 %.
[a] g lignin/100 g OS-A lignin.
[b] g lignin (precipitated + retained in pre-treated solid (OS-A-O) + phenolics)/100 g OS-A lignin.

several putative/partial (lower matching) identification of some of the peaks (data not shown).

3.2.2.2. Lignin precipitation. The Lignin solubilized during organosolv pre-treatment was precipitated with water as described in Section 2.3. Precipitated lignin yield (Eq. (2)) was determined taking into account the lignin content in the raw material (OS-A) and the lignin recovered after organosolv pre-treatment at different experimental conditions (Eq. (3)). Results are shown in Table 6. In the experiments performed at 140 °C the use of sulphuric acid during pre-treatment had a notable effect on the lignin recovered by precipitation, increasing the yield from 7.8 to 21.7 g / 100 g OS-A lignin with the addition of 100 mM sulphuric acid. In contrast, at 165 °C, no significant difference was detected with the addition of acid. The precipitated lignin yield obtained in the experiment performed at 165 °C without acid (20.5 g / 100 g OS-A lignin) was in the range of those obtained at 140 °C with the addition of acid. In the experiments carried out at 190 °C, an increase in precipitated lignin yield was observed when the reaction time rose from 0 to 30 min (from 14.4 to 19.48 g / 100 g OS-A lignin, corresponding from 46.93 g/100 of dissolved lignin). At longer pre-treatment times this yield did not change significantly, with values around 19 g / 100 g OS-A lignin. No direct correlation was found between precipitated lignin yields and the degree of delignification obtained by organosolv pre-treatment (Table 3), possibly, this is because the increased severity of the pre-treatment causes the formation of smaller lignin fragments, which are more difficult to precipitate (Huijgen et al., 2012).

Table 6 also shows the overall lignin recovery, calculated according to Eq. (3) (2.3. section). The results were in the range of 81–97 %. Lower values were obtained for the experiments performed at 190 °C for 30 and 60 min, which were the ones in which higher delignification yields were achieved (Table 3).

These data show that the solubilised lignin recovered by precipitation is not enough to compensate for the lignin loss in the solid and thus corroborate the difficulty to precipitate all the solubilised lignin. A possible alternative in order to try to improve these yields is to distill the ethanol used as a solvent in the pre-treatment. The distillation of the solvent is necessary to recover and reuse it, thus improving the economy of the process. Lignin precipitation would also occur as the ethanolic concentration of the liquor decreases. However, the main drawback of this methodology is that the purity of the precipitated lignin is usually lower since residual sugars remain mixed with the lignin in the precipitate obtained. Depending on the form of lignin valorization, the presence of carbohydrates may not be a problem. Previous studies have shown suitable valorization routes for lignins precipitated from organosolvents to produce activated carbon to be used as a water pollutant adsorbent material (Martín-Sampedro et al., 2019). However, for obtaining other compounds such as adhesives, films and biodegradable polymers, it is necessary to use lignins that present higher purity, which are practically unaltered and less condensed, being partially soluble in many organic solvents (Mesa et al., 2011).

To compare the organosolv delignification yield obtained with conventional delignification methods, an alkaline treatment with NaOH (2 % and 4 %) at 130 °C was also tested. Surprisingly the delignification yields obtained with this treatment were much lower (data not shown), than those obtained with organosolv process. Furthermore, no increase in enzymatic saccharification was observed compared to that obtained without this pre-treatment, showing not only that this methodology was not effective for the biomass here studied, but also that organosolv

Table 7
Glucose concentration, enzymatic saccharification yield and enzymatic hydrolysis (EH) yield obtained from OS-A and OS-A-O.

Sample	Solid yield (%)	Glucose (g/L)	Saccharification yield (%)	EH yield (%)
OS-A	63.1	0.39	–	2.0
OS-A-O_140	79.2	0.99	3.9	3.9
OS-A-O_140-A50	72.3	6.6	24.1	23.7
OS-A-O_140-A100	69.6	8.7	28.1	28.1
OS-A-O_165	64.4	16.3	60.8	52.7
OS-A-O_165-A50	61.4	21.8	84.4	67.1
OS-A-O_190–0	74.2	1.9	6.9	6.9
OS-A-O_190–30	52.4	25.4	95.8	75.9
OS-A-O_190–60	51.8	25.5	100.0	66.7
OS-A-O_190–90	48.3	21.8	98.8	52.7
OS-A-O_190–105	45.7	20.2	100.0	46.3

Deviations < 5 %.

delignification obtained here with OS-A is very interesting. These data contrast with the previously obtained for other biomasses such as *Cistus ladanifer,* which shows higher delignification with NaOH alkaline pre-treatment than with organosolv, achieving a delignification of 76% and an enzymatic yield of 72% at 4% NaOH and 130 °C (Alves-Ferreira et al., 2021).

3.3. Effect of the organosolv pre-treatment conditions on cellulose saccharification

To evaluate the effectiveness of the organosolv pre-treatment on the enzymatic digestibility of cellulose, enzymatic hydrolysis (EH) of the cellulose-enriched solids obtained in the different experimental conditions tested (OS-A-O) was performed. The results are shown in Table 7. For comparison purposes, EH of OS-A used as the raw material in this work is also presented.

The EH yield obtained with OS-A is only 2 %. This EH yield for the non-delignified material is lower than that of other LCB, which is an indication of the high recalcitrance of OS. For example, Martínez-Patiño et al. (2018) obtained an EH yield of 53 % from olive tree biomass after acid pre-treatment.

The low EH yield obtained makes it necessary to perform a second pre-treatment to increase it. In previous work, the steam explosion was used as a second pre-treatment step, saccharification yields increased up to 78 %, and an additional grinding step of the pre-treated solids was necessary (Padilla-Rascón et al., 2020b). In this work, an organosolv pre-treatment has been selected to cause a breakdown of the lignocellulosic structure and swelling of the biomass, increasing the surface contact and improving the enzymatic digestibility (Alves-Ferreira et al., 2021). Other authors have achieved a maximum enzymatic saccharification yield of 81 % with EOP using an organosolv pretreatment (Gómez-Cruz et al., 2022).

In the organosolv tests performed at a lower temperature (140 °C), the saccharification yield of cellulose increased slightly with the addition of sulphuric acid, reaching 28 % (8.7 g/L) with the addition of 100 mM H_2SO_4.

By increasing the temperature to 165 °C the enzymatic digestibility improved, obtaining a saccharification yield of 84.4 % concerning the glucose contained in the OS-A-O g/L for the test carried out with the addition of 50 mM sulphuric acid. However, when the EH yield was calculated, concerning the glucose contained in the OS-A, the result decreased to 67 %, due to the solubilisation of more than 20 % of the cellulose produced under these pre-treatment conditions (Table 3).

In the series of experiments carried out at 190 °C and different pre-treatment times (0, 30, 60, 90 and 105 min), it can be observed that the saccharification yield increased from close to 7–95.8 % from 0 to 30 min. From 60 min of pre-treatment onwards, the saccharification of the cellulose contained in the OS-A-O is complete. On the contrary, a remarkable decrease of the EH yield occurs for times longer than 30 min, due to important cellulose losses. At longer pre-treatment times, more than 30 % of the cellulose is solubilised after 60 min of pre-treatment, reaching about 60 % solubilisation in the experiment carried out at 105 min. A correlation between saccharification and EH yields and solid yields was observed, with EH yields increasing significantly for solid recoveries around 50 % or lower, and saccharification yields decreasing with increasing solid yields from 52 %, which is indicative of higher degradations and solid losses (Table 7).

There is a relationship between the solubilisation of glucose by EH and the degree of delignification of the solid, obtaining higher EH yields from OS-A-O that has had a higher degree of delignification (Table 3). Therefore, we can affirm that the delignification of OS with an organosolv pre-treatment is effective in increasing the solubility of glucose by EH.

In the tested conditions, the highest EH yield (75.9 %) was obtained in the experiment carried out at 190 °C and 30 min. These conditions have therefore been selected for obtaining ethanol from the cellulosic fraction by PSSF, taking into account the recovery of precipitated lignin and the composition of phenolic compounds and their antioxidant capacity (Section 3.2.2.1) were also evaluated for this experiment.

3.4. Ethanol production

Results of ethanol production from the OS-A-O obtained at 190 °C and 30 min by a PSSF process are shown in Fig. 3. It can be seen that the influence of the different enzyme loadings tested (15, 30 and 40 FPU/g OS-A-O) was not very high. Increasing the enzyme dose hardly affects the ethanol concentration at 24 h (they were in the range 29.2–30.3). At 48 h the difference between the ethanol concentrations obtained at the different enzyme loadings was slightly higher (from 38.6 g/L at 15 FPU to 42.5 g/L at 40 FPU). At 72 h, the ethanol concentration with the different enzyme doses tested is 42.3, 47 y 47.2 g/L, con 15, 30 y 40 FPU, respectively. These concentrations correspond to ethanol yields of 55.3, 61.4 and 61.7, respectively, calculated as a percentage of the

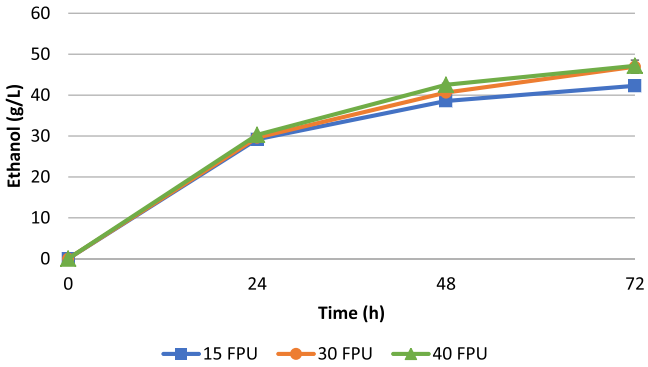


Fig. 3. Ethanol production by PSSF from the solid obtained in the organosolv pre-treatment (OS-A-O) at 190 °C and 30 min with different enzyme loadings 15, 30 and 40 FPU/g OS-A-O.

theoretical maximum that could be obtained from the glucose contained in the OS-A-O (Table 2). These values were similar to that reported by Martínez-Patiño et al. (2017) with olive tree biomass subjected to alkaline delignification with 7 % hydrogen peroxide, in which they obtained 46 g/L ethanol and a yield of 59 %, by simultaneous saccharification and fermentation (SSF) at 20 % solids and 15 FPU/g substrate.

For an ethanol production process to be viable from a distillation efficiency point of view, ethanol concentrations must be above 4 % (w/v). (Koppram et al., 2014). In the present work, this minimum requirement is exceeded at the different enzyme loads tested. The difference obtained at 72 h using 30 or 40 FPU of enzyme is negligible. From the economic point of view, it should be considered the cost of the enzyme versus the benefit obtained in ethanol production, since the cost of enzymes are usually high. Considering profitability, it would be important to determine whether the optimal conditions for obtaining ethanol at 72 h would be those with an enzyme load of 30 FPU/g OS-A, with which a maximum ethanol concentration of 47 g/L was obtained, or whether it would be more profitable to reduce the enzyme load to half (15 FPU/g OS-A-O), with an ethanol concentration drop to 42 g/L. A reduction of time to 48 h could also be of interest. The use of high solid loading is recommended in bioethanol production processes to make the process more economically viable, although a drop in the yield usually occurs. Thus, an economic evaluation would be necessary to determine the most appropriate enzyme dose and fermentation time (Joelsson et al., 2016).

3.5. Furfural production

The xylose-enriched hydrolysate obtained in the acid pretreatment of OS under conditions optimized in previous work (Fig. 1) has a high initial xylose concentration, 65.8 g/L, but also contains high concentrations of inhibitory compounds, mainly acetic acid (15 g/L) and furfural (2.3 g/L), as described in Table 8. As ethanolic fermentation of a similar hydrolysate with the xylose-fermenting strain *Escherichia coli* SL10*0* required extensive detoxification methods to remove inhibitors (unpublished results), in this work, furfural production from OS acid hydrolysates was studied as an alternative approach.

The experimental conditions tested were selected based on previous work (Padilla-Rascón et al., 2021). The catalysts tested were $FeCl_3$ with different concentrations (0–200 mM) and sulphuric acid. The latter was not added as the pre-treated solid already contained it. Sulphuric acid and ferric chloride act as Brønsted and Lewis acids, respectively, in the dehydration reaction of xylose to furfural (Kabbour and Luque, 2020; Padilla-Rascón et al., 2020a). In the present case, sulphuric acid has a dual role, firstly, it acts as a catalyst for the production of xylose from hemicellulose and, subsequently, the acid contained in the hydrolysate acts as Brønsted acid, without the need for its addition to obtain furfural.

The highest furfural concentration produced was 22.7 g/L and it was obtained with 100 mM $FeCl_3$, with a furfural yield of 48.7 % and a xylose conversion of 96.6 % (Fig. 4). At $FeCl_3$ concentrations lower than 100 mM, the furfural produced and the xylose conversion were lower, although a very similar concentration was obtained for 50 mM $FeCl_3$. At higher $FeCl_3$ concentrations (200 mM) the conversion of xylose increased slightly but the furfural yield decreased, therefore, furfural losses were detected, due to degradation by autopolymerisation reactions (Köchermann et al., 2018).

Table 8
Composition of the original acid hydrolysate and the liquors obtained from microwave treatment to obtain furfural at different concentrations of $FeCl_3$ catalyst.

	Acid hydrolysate (g/L)	$FeCl_3$ catalyst concentration (mM)			
		0	50	100	200
Glucose	1.3	0.76	0.32	0.16	0.00
Xylose	65.8	15.5	5.2	2.2	1.5
Galactose	3.5	–	–	–	–
Arabinose	3.7	3.1	2.2	2.0	2.0
Formic acid	0.80	2.2	2.6	2.9	3.2
Acetic acid	15.0	16.8	16.9	16.5	15.9
HMF	0.06	0.19	0.17	0.10	0.09
Furfural	2.3	20.0	22.1	22.7	19.7

Deviations < 5 %.

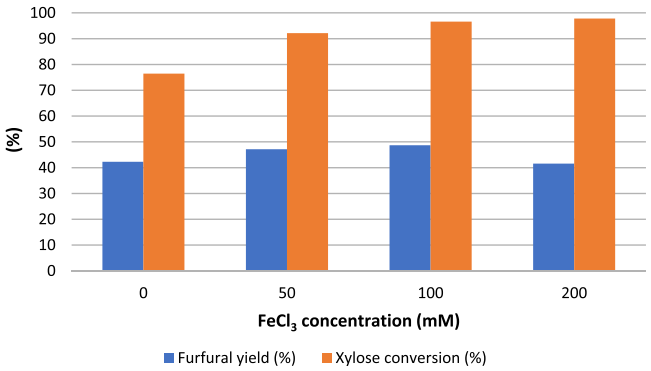


Fig. 4. Concentration and yield of furfural produced and xylose conversion.

In the referred previous work, from an OS hydrolysate with 40.3 g/L xylose concentration, 18 g/L of furfural (63.3 % yield) were obtained, using the same experimental conditions (Padilla-Rascón et al., 2021). In the present work, a lower furfural yield was obtained (48.7 %), but furfural concentration was higher, reaching 22.7 g/L. This can be explained because the liquor used in the present work had a higher xylose concentration (65.8 g/L), which may hinder the production of furfural since it increases its degradation by autopolymerisation. This behaviour has already been observed by other authors, where the lower the xylose concentration of the liquor used, the higher the furfural yields obtained (Fúnez-Núñez et al., 2020).

4. Conclusions

The fractionation strategy proposed in this work allows the full valorisation of the olive stones by enabling the full exploitation of both its polysaccharide fractions and its lignin fraction. The mass balance and yields obtained are shown in detail in Fig. 5.

In the first acid pre-treatment, a hemicellulosic-sugars-rich hydrolysate was obtained, with sugars concentration as high as 73 g/L, equivalent to 22.1 kg hemicellulosic sugars/100 kg OS, the main hemicellulosic sugar being xylose (65.8 g/L, 19.9 kg xylose/100 kg OS). With this acid hydrolysate, a microwave treatment was carried out at 200 °C with $FeCl_3$ 100 mM as a catalyst, obtaining a liquor with a high concentration of furfural (22.5 g/L, 6.8 kg furfural/100 kg OS).

The solid resulting from the first acid pre-treatment was enriched in cellulose and lignin, but still showed low enzymatic saccharification. To increase cellulose digestibility, a organosolv delignification pre-treatment was carried out (190 °C and 30 min), achieving a 49.8 % delignification yield. The organosolv liquor obtained can be used as a source of phenolic compounds. The concentration of TPC obtained was 4.2 kg GAE/ 100 kg OS. The recovery of lignin by precipitation with water was also addressed, enabling to obtain 7.4 kg/100 kg OS. The delignified solid was mainly composed of cellulose and was used to produce ethanol by PSSF with a solid loading of 20 % w/w and an enzyme dosage of 30 FPU, with *S. cerevisiae* as the fermenting microorganism. Ethanol concentration as high as 47 g/L ethanol was obtained (6.2 kg ethanol/100 kg OS) under these conditions.

With the proposed strategy, it was possible to recover 90 % of the sugars contained in the raw OS. From the main sugars, xylose, and glucose, it was possible to obtain furfural and ethanol, with yields of 48.7 % and 61.4 % referring to the xylose and glucose content of the OS, respectively.

This work shows the advantages of a two-step fractionation process (dilute acid hydrolysis followed by organosolv) to obtain multiple

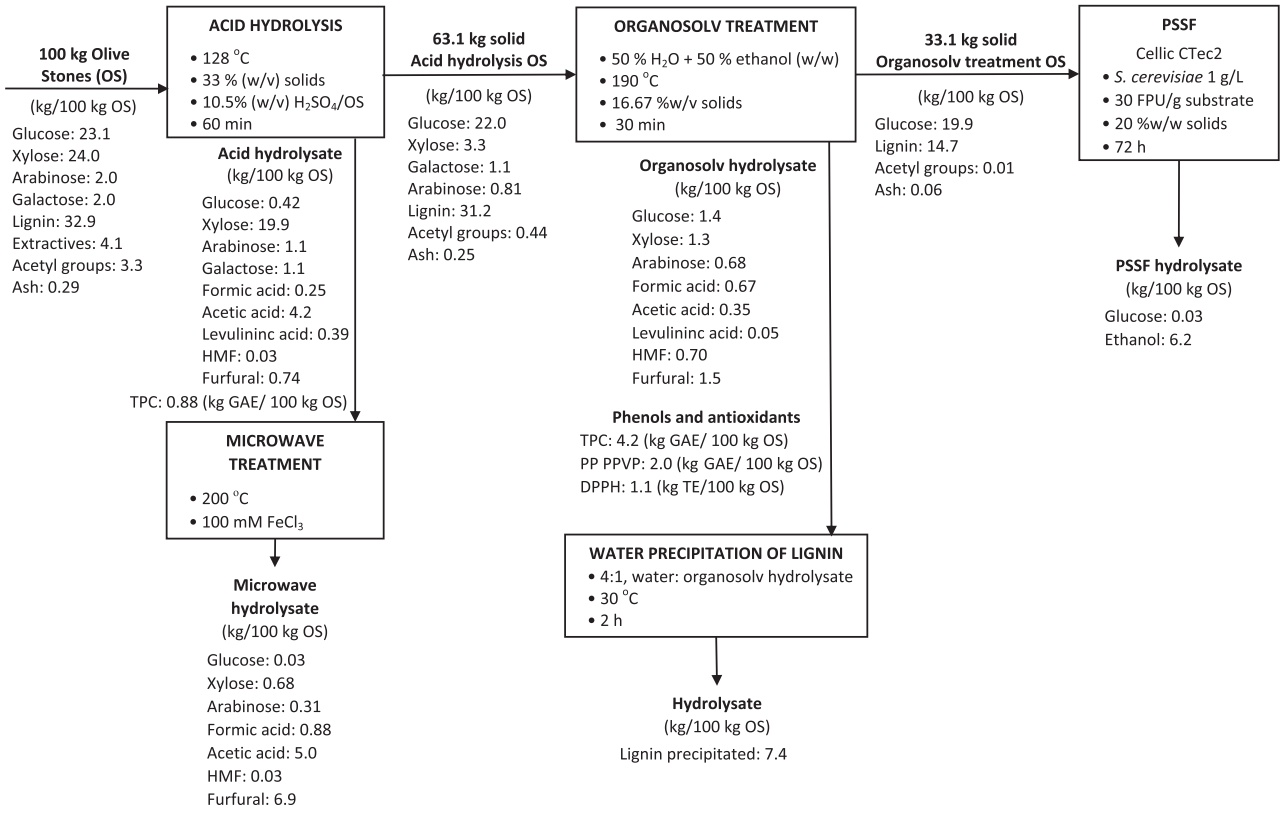


Fig. 5. Mass balance obtained through the integrated fractionation strategy proposed for OS valorisation as carried out under the optimized conditions identified in this work.

biorefinery products from olive stones, i.e., furfural, lignin, phenolic compounds, sugars, and ethanol. Therefore, the full integrated usage of all structural biomass components of olive stone was achieved, and demonstrated to be technologically feasible. Furthermore, there are several significant points in the overall process that can be explored and further optimized based on energy and mass integration systems that could yield significant savings namely on energy utilization and materials (e.g. ethanol and enzymes recycling), that must be further explored in future research. Thus, the data obtained in this work will be an important contribution to techno-economic studies dealing with olives stones-based biorefineries. Future work would be necessary to assess the economic viability and sustainability of the proposed biorefinery.

CRediT authorship contribution statement

CPR conducted the research, and performed the analysis. CPR, FC, LCD, LBR and ER designed the research, performed the data interpretation and discussion and wrote the manuscript. FC, LCD, LBR, ER, and EC supervised the work and revised the manuscript. All authors read and approved the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors want to acknowledge the financial support from Agencia Estatal de Investigación (MICINN, Spain) and Fondo Europeo de Desarrollo Regional, reference project ENE2017–85819-C2–1-R. Carmen Padilla gratefully acknowledges the PhD Fellow grant by Universidad de Jaén (research grant R5/04/2017) and pre-doctoral stay fellowship (Acción 6 para estancias breves de Personal Investigador en Formación encaminadas a la obtención del título de Doctor con mención internacional).

The present work has received funding from Horizon 2020's Research and Innovation Programme under grant agreement number 731101 (BRISK2 project: Biofuels Research Infrastructure for Sharing Knowledge).

This work was partially carried out at the Biomass and Bioenergy Research Infrastructure (BBRI-LISBOA-01–0145-FEDER-022059), which is supported by the Operational Programme for Competitiveness and Internationalization (PORTUGAL2020), by the Lisbon Portugal Regional Operational Programme (Lisboa 2020) and by the North Portugal Regional Operational Programme (Norte2020) under the Portugal 2020 Partnership Agreement, through the European Regional Development Fund (ERDF).

Cellic® CTec2 was kindly provided by Novozymes A/S (Denmark).

The authors thank Belina Ribeiro and Céu Penedo for their laboratory support.

References

Alves-Ferreira, J., Lourenço, A., Morgado, F., Duarte, L.C., Roseiro, L.B., Fernandes, M.C., Pereira, H., Carvalheiro, F., 2021. Delignification of cistus ladanifer biomass by organosolv and alkali processes. Energies 14, 1127. https://doi.org/10.3390/en14041127.

Amendola, D., De Faveri, D.M., Egües, I., Serrano, L., Labidi, J., Spigno, G., 2012. Autohydrolysis and organosolv process for recovery of hemicelluloses, phenolic compounds and lignin from grape stalks. Bioresour. Technol. 107, 267–274. https://doi.org/10.1016/j.biortech.2011.12.108.

Bhatia, S.K., Jagtap, S.S., Bedekar, A.A., Bhatia, R.K., Patel, A.K., Pant, D., Rajesh Banu, J., Rao, C.V., Kim, Y.-G., Yang, Y.-H., 2020. Recent developments in pretreatment technologies on lignocellulosic biomass: effect of key parameters, technological improvements, and challenges. Bioresour. Technol. 300, 122724 https://doi.org/10.1016/j.biortech.2019.122724.

Chen, H., Zhao, J., Hu, T., Zhao, X., Liu, D., 2015. A comparison of several organosolv pretreatments for improving the enzymatic hydrolysis of wheat straw: Substrate digestibility, fermentability and structural features. Appl. Energy 150, 224–232. https://doi.org/10.1016/j.apenergy.2015.04.030.

Contreras, M., del, M., Romero, I., Moya, M., Castro, E., 2020. Olive-derived biomass as a renewable source of value-added products. Process Biochem 97, 43–56. https://doi.org/10.1016/j.procbio.2020.06.013.

Díaz, M.J., Huijgen, W.J.J., van der Laan, R.R., Reith, J.H., Cara, C., Castro, E., 2011. Organosolv pretreatment of olive tree biomass for fermentable sugars. Holzforschung 65. https://doi.org/10.1515/hf.2011.030.

Doménech, P., Duque, A., Higueras, I., Iglesias, R., Manzanares, P., 2020. Biorefinery of the olive tree—production of sugars from enzymatic hydrolysis of olive stone pretreated by alkaline extrusion. Energies 13, 4517. https://doi.org/10.3390/en13174517.

Ferreira, J.A., Taherzadeh, M.J., 2020. Improving the economy of lignocellulose-based biorefineries with organosolv pretreatment. Bioresour. Technol. 299, 122695 https://doi.org/10.1016/j.biortech.2019.122695.

Fúnez-Núñez, I., García-Sancho, C., Cecilia, J.A., Moreno-Tost, R., Serrano-Cantador, L., Maireles-Torres, P., 2020. Recovery of pentoses-containing olive stones for their conversion into furfural in the presence of solid acid catalysts. Process Saf. Environ. Prot. 143, 1–13. https://doi.org/10.1016/j.psep.2020.06.033.

Gómez-Cruz, I., Contreras, M., del, M., Carvalheiro, F., Duarte, L.C., Roseiro, L.B., Romero, I., Castro, E., 2021. Recovery of bioactive compounds from industrial exhausted olive pomace through ultrasound-assisted extraction. Biology 10, 514. https://doi.org/10.3390/biology10060514.

Gómez-Cruz, I., Romero, I., Contreras, M., del, M., Labidi, J., Hernández-Ramos, F., Roseiro, L.B., Duarte, L.C., Castro, E., Carvalheiro, F., 2022. Combined extraction and ethanol organosolv fractionation of exhausted olive pomace for bioactive compounds. Adv. Sustain. Syst. 2100361. https://doi.org/10.1002/adsu.202100361.

Huijgen, W.J.J., Reith, J.H., den Uil, H., 2010. Pretreatment and fractionation of wheat straw by an acetone-based organosolv process. Ind. Eng. Chem. Res. 49, 10132–10140. https://doi.org/10.1021/ie101247w.

Huijgen, W.J.J., Smit, A.T., Reith, J.H., Uil, H. den, 2011. Catalytic organosolv fractionation of willow wood and wheat straw as pretreatment for enzymatic cellulose hydrolysis. J. Chem. Technol. Biotechnol. 86, 1428–1438. https://doi.org/10.1002/jctb.2654.

Huijgen, W.J.J., Smit, A.T., de Wild, P.J., den Uil, H., 2012. Fractionation of wheat straw by prehydrolysis, organosolv delignification and enzymatic hydrolysis for production of sugars and lignin. Bioresour. Technol. 114, 389–398. https://doi.org/10.1016/j.biortech.2012.02.143.

Joelsson, E., Erdei, B., Galbe, M., Wallberg, O., 2016. Techno-economic evaluation of integrated first- and second-generation ethanol production from grain and straw. Biotechnol. Biofuels 9, 1. https://doi.org/10.1186/s13068-015-0423-8.

Jönsson, L.J., Alriksson, B., Nilvebrant, N.-O., 2013. Bioconversion of lignocellulose: inhibitors and detoxification 10. https://doi.org/10.1186/1754–6834-6–16.

Kabbour, M., Luque, R., 2020. Furfural as a platform chemical. Biomass, fuels, Biochemicals. Elsevier, pp. 283–297. https://doi.org/10.1016/B978-0-444-64307-0.00010-X.

Kim, D., 2018. Physico-chemical conversion of lignocellulose: inhibitor effects and detoxification strategies: a mini review. Molecules 23, 309. https://doi.org/10.3390/molecules23020309.

Köchermann, J., Mühlenberg, J., Klemm, M., 2018. Kinetics of hydrothermal furfural production from organosolv hemicellulose and D -Xylose. Ind. Eng. Chem. Res. 57, 14417–14427. https://doi.org/10.1021/acs.iecr.8b03402.

Koppram, R., Tomás-Pejó, E., Xiros, C., Olsson, L., 2014. Lignocellulosic ethanol production at high-gravity: challenges and perspectives. Trends Biotechnol. 32, 46–53. https://doi.org/10.1016/j.tibtech.2013.10.003.

López-Linares, J.C., Moya, M., Peláez, L., Ruiz, E., Romero, I., 2018. Valorización de hoja de almazara en el contexto de una biorrefinería. Extracción acuosa en autoclave. Afinidad 76, 93–102.

López-Linares, J.C., Gómez-Cruz, I., Ruiz, E., Romero, I., Castro, E., 2020. Production of ethanol from hemicellulosic sugars of exhausted olive pomace by escherichia coli. Processes 8, 533. https://doi.org/10.3390/pr8050533.

Manzanares, P., Ruiz, E., Ballesteros, M., Negro, M.J., Gallego, F.J., López-Linares, J.C., Castro, E., 2017. Residual biomass potential in olive tree cultivation and olive oil industry in Spain: valorization proposal in a biorefinery context. Span. J. Agric. Res. 15, e0206 https://doi.org/10.5424/sjar/2017153-10868.

Martínez-Patiño, J.C., Ruiz, E., Romero, I., Cara, C., López-Linares, J.C., Castro, E., 2017. Combined acid/alkaline-peroxide pretreatment of olive tree biomass for bioethanol production. Bioresour. Technol. 239, 326–335. https://doi.org/10.1016/j.biortech.2017.04.102.

Martínez-Patiño, J.C., Ruiz, E., Cara, C., Romero, I., Castro, E., 2018. Advanced bioethanol production from olive tree biomass using different bioconversion schemes. Biochem. Eng. J. 137, 172–181. https://doi.org/10.1016/j.bej.2018.06.002.

Martín-Sampedro, R., Eugenio, M.E., Fillat, Ú., Martín, J.A., Aranda, P., Ruiz-Hitzky, E., Ibarra, D., Wicklein, B., 2019. Biorefinery of lignocellulosic biomass from an elm clone: production of fermentable sugars and lignin-derived biochar for energy and environmental applications. Energy Technol. 7, 277–287. https://doi.org/10.1002/ente.201800685.

Mesa, L., González, E., Cara, C., González, M., Castro, E., Mussatto, S.I., 2011. The effect of organosolv pretreatment variables on enzymatic hydrolysis of sugarcane bagasse. Chem. Eng. J. 168, 1157–1162. https://doi.org/10.1016/j.cej.2011.02.003.

Miliotti, E., Dell'Orco, S., Lotti, G., Rizzo, A., Rosi, L., Chiaramonti, D., 2019. Lignocellulosic ethanol biorefinery: valorization of lignin-rich stream through hydrothermal liquefaction. Energies 12, 723. https://doi.org/10.3390/en12040723.

Moniz, P., João, L., Duarte, L.C., Roseiro, L.B., Boeriu, C.G., Pereira, H., Carvalheiro, F., 2015. Fractionation of hemicelluloses and lignin from rice straw by combining autohydrolysis and optimised mild organosolv delignification. BioResources 10, 2626–2641. https://doi.org/10.15376/biores.10.2.2626-2641.

Mosier, N., 2005. Features of promising technologies for pretreatment of lignocellulosic biomass. Bioresour. Technol. 96, 673–686. https://doi.org/10.1016/j.biortech.2004.06.025.

Padilla-Rascón, C., Romero-García, J.M., Ruiz, E., Castro, E., 2020a. Optimization with response surface methodology of microwave-assisted conversion of Xylose to Furfural. Molecules 25, 3574. https://doi.org/10.3390/molecules25163574.

Padilla-Rascón, C., Ruiz, E., Romero, I., Castro, E., Oliva, J.M., Ballesteros, I., Manzanares, P., 2020b. Valorisation of olive stone by-product for sugar production using a sequential acid/steam explosion pretreatment. Ind. Crops Prod. 148, 112279 https://doi.org/10.1016/j.indcrop.2020.112279.

Padilla-Rascón, C., Romero-García, J.M., Ruiz, E., Romero, I., Castro, E., 2021. Microwave-assisted production of furfural from the hemicellulosic fraction of olive stones. Process Saf. Environ. Prot. 152, 630–640. https://doi.org/10.1016/j.psep.2021.06.035.

Pan, X., Arato, C., Gilkes, N., Gregg, D., Mabee, W., Pye, K., Xiao, Z., Zhang, X., Saddler, J., 2005. Biorefining of softwoods using ethanol organosolv pulping: Preliminary evaluation of process streams for manufacture of fuel-grade ethanol and co-products. Biotechnol. Bioeng. 90, 473–481. https://doi.org/10.1002/bit.20453.

Romero, I., López-Linares, J.C., Delgado, Y., Cara, C., Castro, E., 2015. Ethanol production from rape straw by a two-stage pretreatment under mild conditions. Bioprocess Biosyst. Eng. 38, 1469–1478. https://doi.org/10.1007/s00449-015-1389-4.

Romero-García, J.M., Niño, L., Martínez-Patiño, C., Álvarez, C., Castro, E., Negro, M.J., 2014. Biorefinery based on olive biomass. state of the art and future trends. Bioresour. Technol. 159, 421–432. https://doi.org/10.1016/j.biortech.2014.03.062.

Roseiro, L.B., Tavares, C.S., Roseiro, J.C., Rauter, A.P., 2013. Antioxidants from aqueous decoction of carob pods biomass (Ceretonia siliqua L.): optimisation using response surface methodology and phenolic profile by capillary electrophoresis. Ind. Crops Prod. 44, 119–126. https://doi.org/10.1016/j.indcrop.2012.11.006.

Ruiz, E., Romero-García, J.M., Romero, I., Manzanares, P., Negro, M.J., Castro, E., 2017. Olive-derived biomass as a source of energy and chemicals. Biofuels Bioprod. Bioref. 11, 1077–1094. https://doi.org/10.1002/bbb.1812.

Scopel, E., Rezende, C.A., 2021. Biorefinery on-demand: modulating pretreatments to recover lignin, hemicellulose, and extractives as co-products during ethanol production. Ind. Crops Prod. 163, 113336 https://doi.org/10.1016/j.indcrop.2021.113336.

Singleton, V.L., Rossi, J.A., 1965. Colorimetry of total phenolics with phosphomolybdic-phosphotungstic acid reagents. Am. J. Enol. Vitic. 16, 144.

Sluiter, A., Hames, B., Ruiz, R., Scarlata, C., Sluiter, J., Templeton, D., Crocker, D., 2012. Determination of Structural Carbohydrates and Lignin in Biomass, Revised August 2012. ed. National Renewable Energy Laboratory, Golden, Colorado.

Taherzadeh, M., Karimi, K., 2008. Pretreatment of lignocellulosic wastes to improve ethanol and biogas production: a review. Int. J. Mol. Sci. 9, 1621–1651. https://doi.org/10.3390/ijms9091621.

Wildschut, J., Smit, A.T., Reith, J.H., Huijgen, W.J.J., 2013. Ethanol-based organosolv fractionation of wheat straw for the production of lignin and enzymatically digestible cellulose. Bioresour. Technol. 135, 58–66. https://doi.org/10.1016/j.biortech.2012.10.050.

Yang, H., Yoo, C.G., Meng, X., Pu, Y., Muchero, W., Tuskan, G.A., Tschaplinski, T.J., Ragauskas, A.J., Yao, L., 2020. Structural changes of lignins in natural Populus variants during different pretreatments. Bioresour. Technol. 295, 122240 https://doi.org/10.1016/j.biortech.2019.122240.

Zakzeski, J., Bruijnincx, P.C.A., Jongerius, A.L., Weckhuysen, B.M., 2010. The catalytic valorization of lignin for the production of renewable chemicals. Chem. Rev. 110, 3552–3599. https://doi.org/10.1021/cr900354u.

Zhang, K., Pei, Z., Wang, D., 2016. Organic solvent pretreatment of lignocellulosic biomass for biofuels and biochemicals: a review. Bioresour. Technol. 199, 21–33. https://doi.org/10.1016/j.biortech.2015.08.102.

Zhao, X., Cheng, K., Liu, D., 2009. Organosolv pretreatment of lignocellulosic biomass for enzymatic hydrolysis. Appl. Microbiol. Biotechnol. 82, 815–827. https://doi.org/10.1007/s00253-009-1883-1.